MIL-T-25121A (USAF)
21 April 1964

Superseding
MIL-T-25121 (USAF)
11 April 1956

MILITARY SPECIFICATION

TRAINING AID, NIGHT VISION, TYPE MP-1

1. SCOPE

1.1 This specification covers one type of night vision training aid, designated Type MP-1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-530	Plywood, Flat Panel
TT-S-711	Stain, Oil Type, Wood, Interior
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood Cleated
PPP-B-601	Boxes, Wood Cleated, Plywood
PPP-B-621	Boxes, Wood Nailed and Lock-corner
PPP-B-636	Boxes, Fiberboard
PPP-C-843	Cushioning Material, Cellulosic

Military

JAN-P-100	Packaging and Packing For Overseas Shipment; General Specification
MIL-B-107	Boxes, Wood, Wirebound, "Overseas Type"
MIL-P-116	Preservation, Methods Of
MIL-E-5558	Enamel; Wrinkle-Finish, For Aircraft Use
MIL-L-7178	Lacquer; Cellulose Nitrate, Gloss For Aircraft Use
MIL-H-7298	Handbooks; Standard Commercial Equipment
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-D-70327	Drawings, Engineering & Associated Lists

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property

FSC 6910

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MIL-STD-143	Specifications and Standards; Order of Precedence For The Selection Of
MS-33586	Metals; Definition of Dissimilar

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing.

3.2 Components.- The training aid shall consist of the following:

ITEM NR	DESCRIPTION	SEE REQUIREMENT
1	Projection System	3.7
2	Projection Changing System	3.8
3	Cooling System	3.9
4	Case (Housing)	3.10
5	Accessory Equipment	3.11

3.3 Selection of Specifications and Standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial Parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard Parts.- With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials

3.4.1 Fungus-Proof Materials.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

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3.4.2 Metals.- Metals shall be of the corrosion-resistant type or suitable treated to resist corrosion due to fuels, salt spray, or atmospheric conditions as may be encountered in storage or normal service.

3.4.2.1 Dissimilar Metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS-33586.

3.5 Design and Construction.- The training aid shall be designed and constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 Maintenance.- The training aids shall be constructed so that adjustments and repairs can be made easily by the personnel of operating units and overhaul bases without the use of special tools.

3.5.2 The training aid shall be designed and constructed so that, as nearly as possible, no light will be transmitted from the projector to the outside of the housing case which will interfere with projection of the demonstrations in totally darkened rooms.

3.5.3 Power Source.- The training aids shall be designed to operate from a 115V, 60-cycle, a-c power source.

3.6 Performance.- The training aids shall be capable of performing demonstrations of the following aspects of night vision:

a. Dark adaption.

b. Off-center vision.

c. Decline and loss of color vision.

d. Purkinje phenomenon.

e. Decline of visual acuity.

f. Increase of contrast threshold

g. Usefulness of scanning methods.

h. Effect of short-time glare and recovery after glare.

i. Local adaption.

j. Autokinetic illusion and prevention by use of multiple lamp installation.

3.7 Projection System.- The projection system shall consist of the following:

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3.7.1 Projector.- The projector shall be constructed of a lightweight material and shall be designed to facilitate the incorporation of a focusing lamp mount or a fixed-focus lamp mount utilizing a prefocusing bulb, condensing reflector mirror (S on figure 1), condensing lenses (C on Figures 1 and 2), and a cooling fan in the area of the lamp housing.

3.7.2 Light Source.- The light source shall be a 115V projection bulb (A on Figure 1) for the smallest type of filament that will, when projecting a clear beam of light through the unhindered projection path, produce an even illumination of not less than 0.15 foot candles nor more than 0.30 foot candles at a distance of 330 cm from the projection lens in a totally darkened room. When used in the same manner and utilizing filters of densities 1, 2, 3, and 4, the illumination shall measure 10, 1, 0.1, .01 per cent of the foregoing amounts, respectively. The projection lamp shall be provided with a variable resistor to enable the operator to control and vary the intensity of the light. The projected image through the translucent density filters will be variable throughout a continous range.

3.7.3 Optical arrangement.- The optical arrangement shall consist of a concave reflector (B on Figure 1) located at the rear of the projector in back of the projection bulb, condensing lenses as indicated by G on Figure 1, and wide angle lenses (R on Figures 1 and 3).

3.7.4 Base Mount.- The base mount shall be constructed of lightweight metal and shall support the projector portion of the training aid and the turret shafts. The turret shafts shall be positioned above the projector as shown on Figure 2. The base shall be attached to the rear panel of the case so as to allow the base and its affixed parts to be withdrawn to the rear when the back panel is removed from the case. The base shall rest on the bottom of the training aid case between two guide rails located within a permanently positioned and prescribed area. Suitable wing-nut locking screws shall be provided to lock the base in position.

3.8 Projection changing system.- The projection changing system shall include the apparatus for introducing various density filters and images into the projection path so that the amount of light reaching the screen (S on Figure 1) may be varied and the areas of relative luminance illustrated.

3.8.1 Density Filter Changing System.- The turret changer shall be located immediately forward and adjacent to the most forward condensing lens (C on Figure 1) and shall not impose any friction resistance to the condensing lens. The turret (D on Figure 1) shall consist of a disc of lightweight metal approximately 9-1/2 inches in diameter and shall have five cutout areas capable of accepting 2-1/2- by 2-1/2-inch glass density filters. The areas

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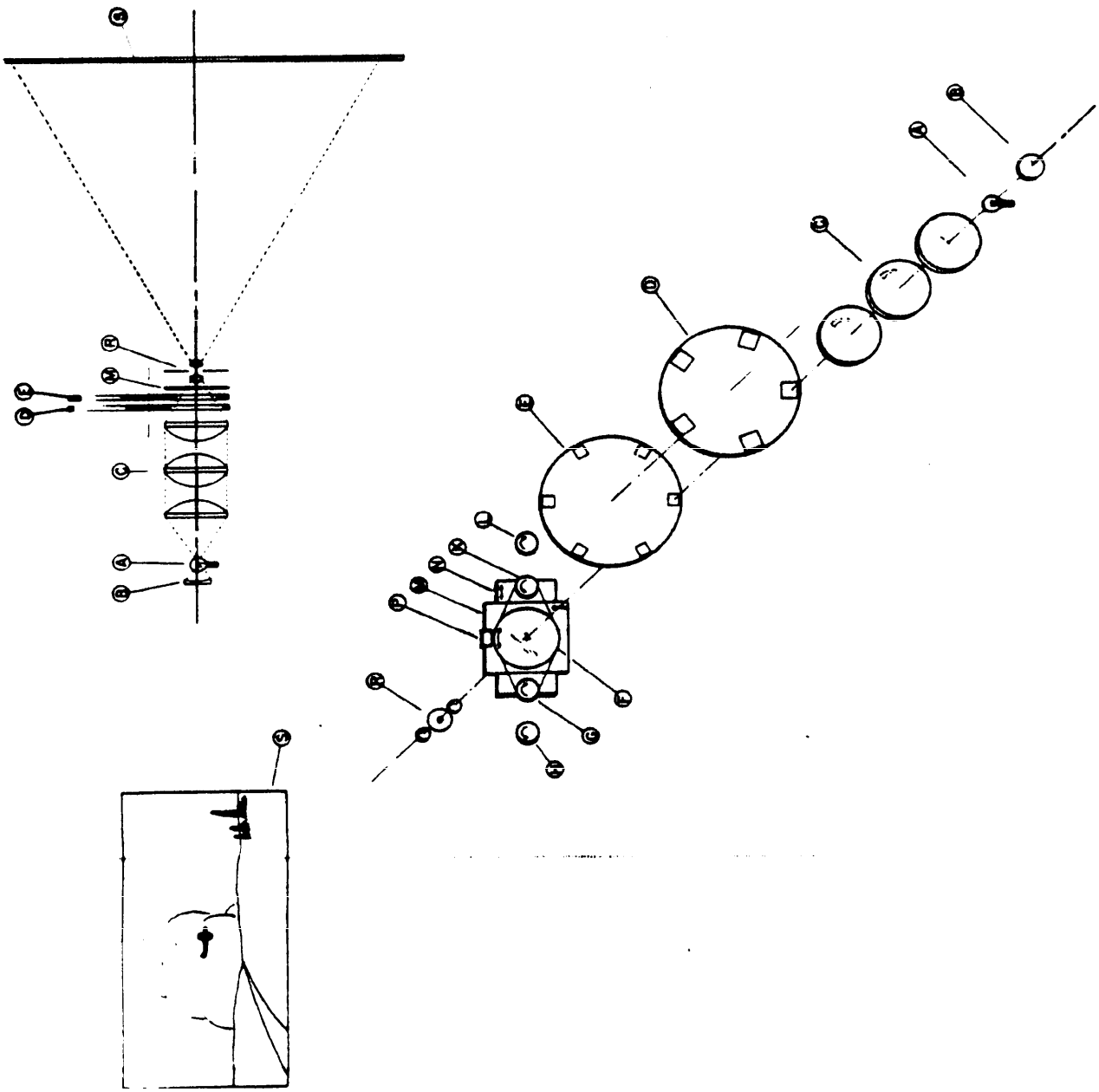


FIGURE 1 CONDENSING LENSES

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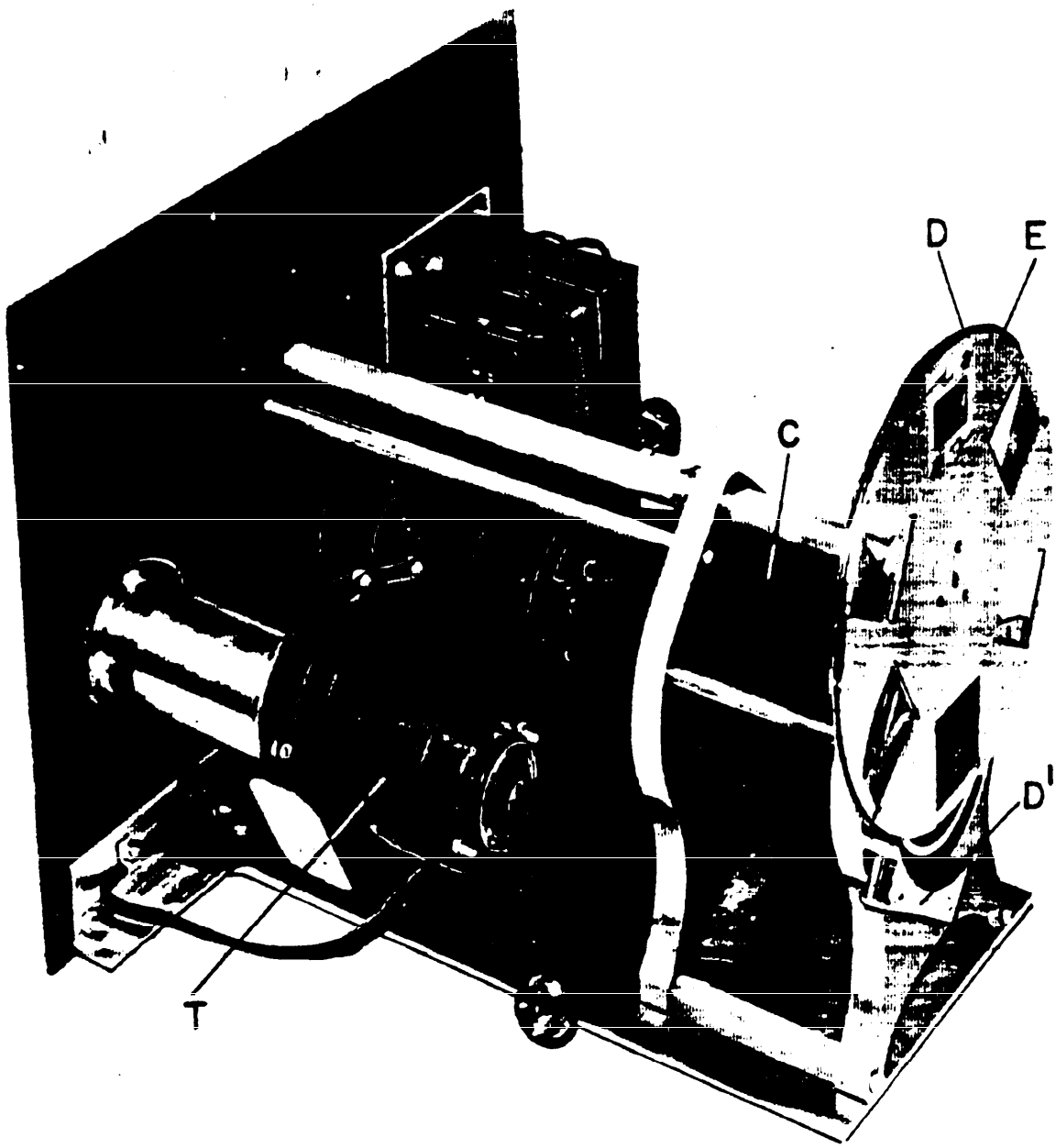


FIGURE 2 PROJECTOR

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shall be positioned around the surface of the turret disc so as to allow any of the filters to be introduced into the projection path of the projection system. The filter turret shall be positioned by adjusting knob U, as shown on figure 4, located at the terminus of the shaft which extends through the rear panel. The shaft shall be affixed to filter turret D as shown on Figure 2.

3.8.2 Slide Changing System.- The slide changing system shall be positioned immediately forward and adjacent to the density filter turret as shown on Figure 2 and shall not impose any friction resistance to the density filter turret. The turret (F on Figures 1 and 2) shall consist of a disc approximately 9-1/2 inches in diameter and shall have six cutout areas capable of accepting 2- by 2-inch paper and glassmounted slides which shall conform to rectangular dimensions of 35 mm. The areas shall be positioned around the surface of the turret disc so as to allow any of the landscape picture slides to be introduced into the projection path so that any slide image may be combined with, or without, a density filter (see D and E on Figure 1). Positioning of the various slides in the projection path shall be accomplished by adjusting knob V (as shown on Figure 4) located at the terminus of the shaft which extends through the rear panel. The shaft shall be affixed to the slide turret. The shaft shall be contained within and act independently of the shaft affixed to the density filter turret.

3.8.3 Image Changing System.- The image changing system shall be positioned forward and immediately adjacent to the slide changing turret (E on Figures 1, 2, and 4). It shall impose no friction resistance to the slide changing system. It shall depict a silhouette one mm in length representing a modern aircraft which shall be transported in proper direction across the field of view.

3.8.3.1 Operation of Image Changing System.- The image changing system shall be operated by mechanical means in such manner that the silhouettes are transported across the projection path on a horizontal plane. The horizontal linear path shall be reversible 180°. The time required to traverse the projection path shall be 90 seconds. Provisions shall be made by which the silhouette can be removed for indefinite periods from the projection path. The aircraft silhouette shall be capable of crossing a projected image from the same direction every 200 seconds or from opposite directions once every 100 seconds. The noise, if any, made by the silhouette changing directions shall be inaudible or suitably masked by other noise cues so that an observer will be unable to detect audibly the change in direction of the silhouette. The lens shall be approximately 28 mm focal length, wide angle type, with a speed of approximately f/16.

3.9 Cooling System.- A cooling fan shall be provided for directing air onto the projection lamp. Cool air shall be conducted from the outside of the training aid through a light trap into the fan by means of a duct. A second duct shall conduct the air from the fan onto the projection lamp. A temperature of

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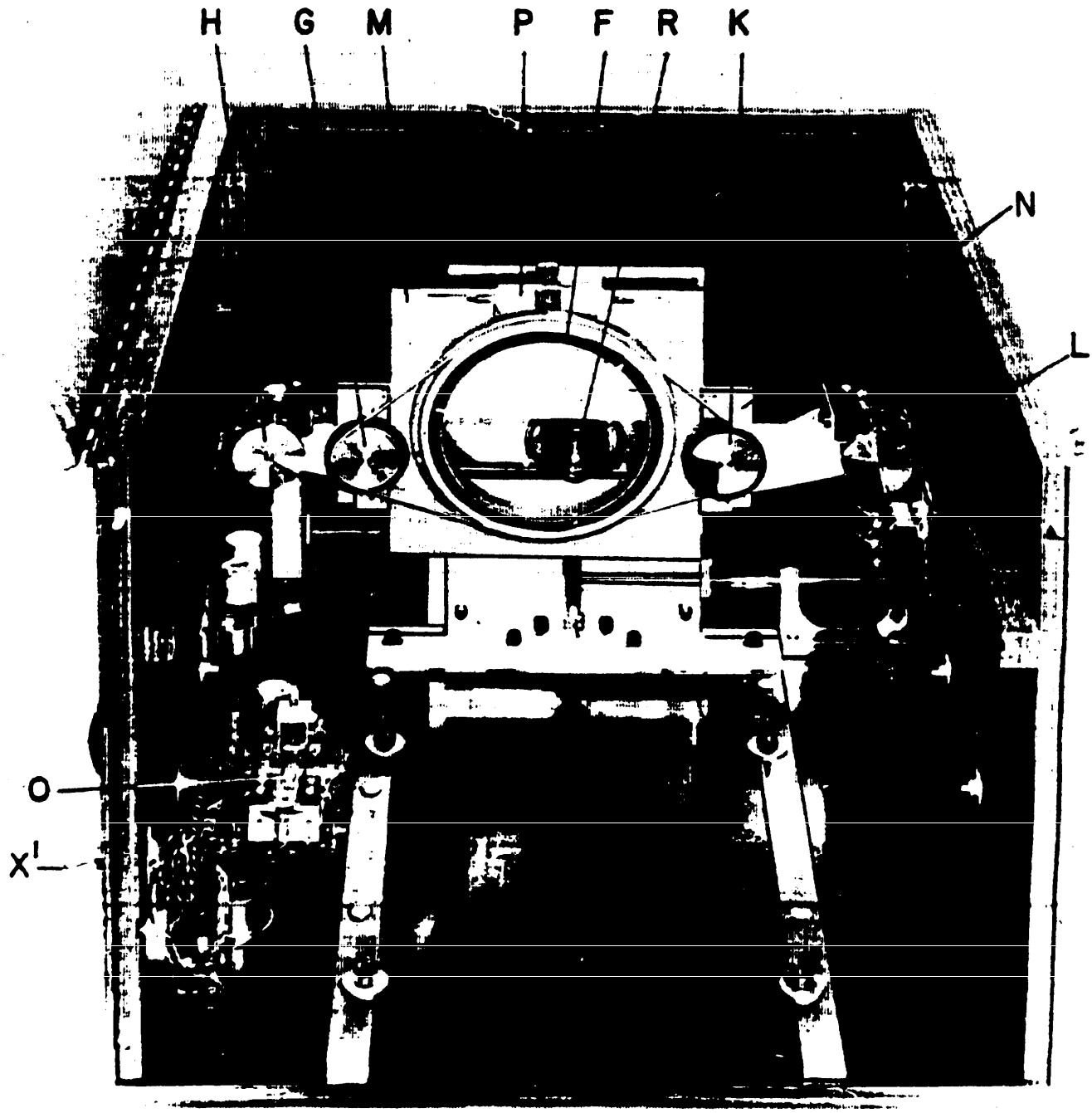


FIGURE 3 SECT VIEW OF PROJECTOR

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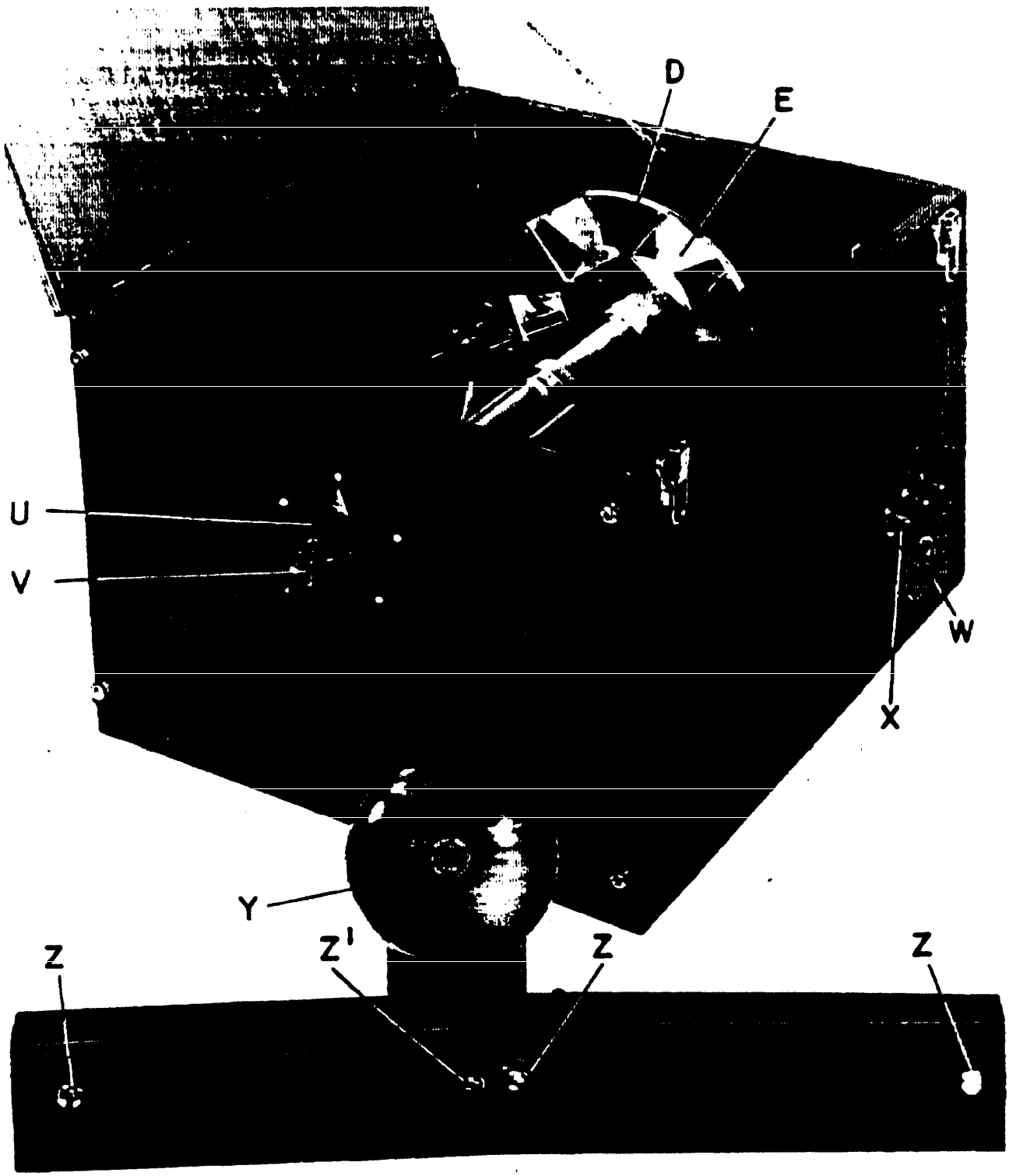


FIGURE 4 FILTER TURRET

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not more than 30° above ambient room temperature shall be maintained as measured at the emulsion side of the transparency.

3.9.1 Cooling Ports.- The training aid shall be provided with vents to efficiently cool the inner area by convection. The ports shall be provided with light traps.

3.10 Case (Housing).- The case which provides housing for the projection system shall be constructed of 5/8-inch plywood, rotary cut, grade 1, in accordance with NN-P-530. The external maximum configurations of the case shall measure 14-1/2 by 20 inches. A hinged cover of 5/8-inch plywood shall be affixed in a manner which shall allow exposure of the complete training aid interior for maintenance purposes. The rear panel of the case, opposite the side containing the lens, shall be constructed of lightweight metal. The metal side shall be capable of supporting the rear of the slide and filter changing shafts, lamp house blower, base mount, indicator dials, switches, and the power connections, as well as being capable of bearing its stress or the component of the case. The metal side shall be detachable from the case and shall be fastened in place by a sufficient number of screws to maintain a tight stable case. A tray made of a lighter weight plywood shall form a part of the case. The tray shall nest in the upper portion of the case and shall be removable. The tray shall have at least three pins capable of containing the accessory equipment. Means shall be provided for cushioning the glass accessory components to be carried in the tray. A series of switches, located on the rear panel shall control the horizontal movement switches, projection light and fan, glare source, and autokinetic demonstration light switches. Flush or countersunk type metal handles of a size and strength which shall support the weight of the complete training aid, when lids are secured, shall be provided.

3.11 Accessory Equipment

3.11.1 Film Slides.- A set of four paper-mounted 35 mm positive color-transparent slides shall be provided. Each slide shall depict a different landscape view and at least two slides shall encompass an airfield area. A fifth slide shall be a hypothetical picture of a horizon illustrating various silhouette patterns of typical occurring structures, as shown on Figure 5.

3.11.2 Light Bar Assembly

3.11.2.1 Glare Bar.- A glare bar assembly shall be constructed as shown on Figure 4. The assembly shall have provisions for accepting a reflector-type photo floodlamp to be used as a glare source.

3.11.2.2 Autokinetic Light Bar.- The autokinetic light bar shall be constructed as shown on Figure 4. The light bar shall consist of two bars hinged together so that when extended they

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appear as one. The bars shall contain four lights as shown on Figure 4. The glare light shall be controlled by toggle switches at the side of the training aid case. The switches controlling the automatic light bar shall be such that the central lighting of the three lights may be switched on independently or in unison with the two remaining lights.

3.11.3 Cables

3.11.3.1 Main Power Cable.- The main power cable shall be approximately 30 feet in length and shall consist of two-conductor rubber-covered wire. Each conductor shall be capable of carrying the current necessary for the operation of the training aid. Suitable strain reliefs shall be incorporated at each end of the cable. One end shall terminate in a two-prong male plug for attachment to the power source and the other end shall terminate in a female receptacle for attachment to the main lead at the training aid case.

3.11.3.1.1 Bar Assembly Cable.- A cable, 15 feet in length, conforming to the requirements specified in 3.11.3.1 shall be provided for connecting the training aid to the light bar assembly.

3.12 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements MIL-D-70327 shall govern changes in the manufacturer's part numbers.

3.13 Finishes and Protective Coatings

3.13.1 Plywood.- Plywood shall be stained with a medium brown mahogony stain conforming to TT-S-711 followed by two coats of clear lacquer conforming to MIL-L-7178. The lacquer coating shall be rubbed to a dull luster.

3.13.2 Metal.- Metal parts, when practicable, shall be finished with black wrinkle enamel conforming to MIL-E-5558.

3.14 Identification of Product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.15 Workmanship.- Workmanship shall be in accordance with highgrade manufacturing practice for this type of equipment.

3.16 Lecture and Maintenance Data.- Two copies of the commercial lecture and maintenance data in accordance with MIL-H-7298 should be provided with each training aid.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase

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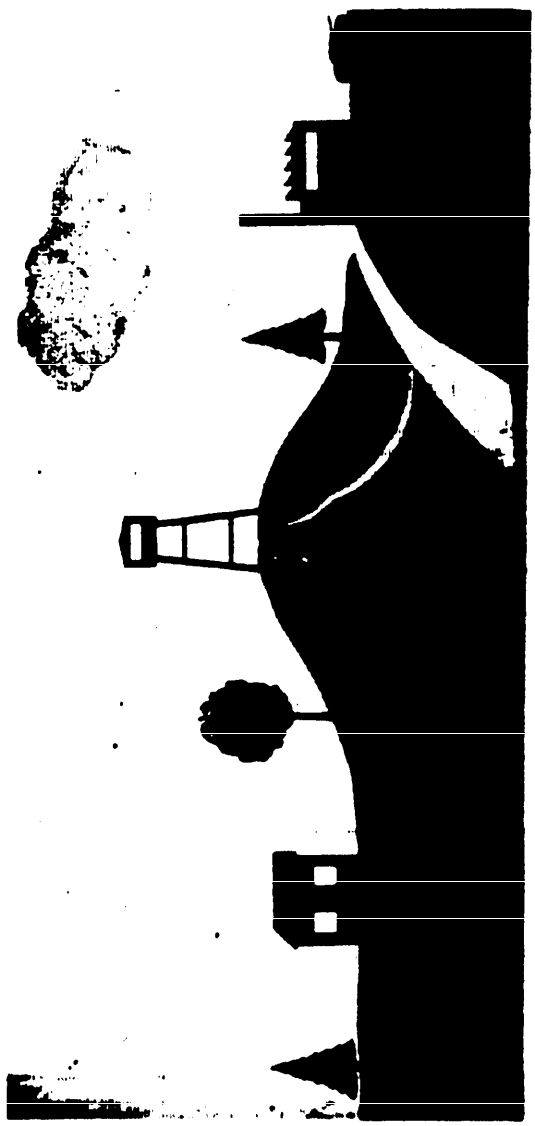


FIGURE 5 SILHOUETTE PATTERN

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order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Tests.- The inspection and testing of the training aid shall be classified as follows:

a. Acceptance tests . . . . . See 4.3

b. Preproduction testing . . . . . See 4.5

4.3 Acceptance Tests.- Acceptance tests shall consist of the following individual tests:

4.3.1 Individual Tests.- Each training aid shall be subjected to the following tests as described under 4.4:

a. Examination of product.

b. Operational.

4.4 Test Methods

4.4.1 Examination of Product.- The training aid shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.4.2 Operation.- The training aid shall be operated through 50 complete cycles. Following this period, the training aid shall be inspected for loose connections, binding and worn parts, and undue friction. The electrical units shall be checked for proper operation.

4.4.3 Projection Test.- With the projector set at a distance of 330 cm from the screen and without a slide or filters in the projector, the projection lamp shall project a clear light upon the screen. The illumination on the screen shall measure 0.15 to 0.30 foot candles at the center and shall measure at least 45 per cent as great illumination at the extremities of the picture with even gradation of light intensity from the center of the screen to the extremities. A series of 1-inch squares shall be marked on the four corners of the screen, the center, and at the corners of a hypothetical rectangle one-half the linear dimension of the total projection. By the same procedure, measurements shall be taken utilizing filters 1, 2, 3, and 4. The measurements shall be as specified in 3.7.2 (1 nit equals 1 candle per m^2).

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4.4.4 Complete Training Procedure Test.- The training aid shall be run through a complete training procedure demonstrating effectively to a group of not less than five viewers, one of whom shall be familiar with night vision problems and phenomenon specified in 3.6.

4.4.5 Cooling System.- The cooling system shall be capable of maintaining a temperature of not more than 30° above ambient room temperature for a period of not less than 3 continuous hours of operation of the training aid. The temperature shall be measured at the emulsion side of the transparency by utilization of a suitable thermocouple.

4.5 Preproduction Testing

4.5.1 Preproduction Test Sample.- The preproduction test sample shall consist of one training aid representative of the production equipment. The training aid shall be tested at a laboratory designated by the procuring activity or, when so stated in the contract, at the contractor's plant under the supervision of the procuring activity.

4.5.2 Preproduction tests.- Preproduction tests shall consist of all tests described under 4.4.

5. PREPARATION FOR DELIVERY

5.1 Application.- The requirements of section 5 apply only to direct purchases by or direct shipment to the Government.

5.2 Preservation and Packaging

5.2.1 Immediate Use (For Domestic Shipment With Immediate Use at Initial Destination).- Night vision training aids shall be protected against corrosion and deterioration in accordance with good commercial practice.

5.2.2 Domestic Shipment With Storage For Overseas Shipment.- Night vision training aids shall receive protection in accordance with MIL-P-116, method II. Points of contact between sharp edges and corners likely to cause puncture of the barrier material shall be cushioned with material conforming to PPP-C-843.

5.3 Packing

5.3.1 Immediate Use (For Domestic Shipment with Immediate Use at Initial Destination).- Packages which require overpacking for acceptance by the carrier shall be packed in commercial exterior-type shipping containers in a manner that will insure safe transportation, at the lowest rate, to the point of delivery. Containers shall meet Consolidated Freight Classification Rules or regulations of other common carriers, as applicable to the mode of transportation.

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5.3.2 Domestic Shipment.- Night vision training aids packaged as specified in 5.2.2 shall be packed in domestic-type exterior containers conforming to PPP-B-591, PPP-B-621, PPP-B-636, PPP-B-585, PPP-B-601, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. Insofar as practical, exterior containers shall be of uniform shape and size and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Strapping and closure shall be in accordance with the appendix of the applicable container specification. When fiberboard containers are used, the fiberboard shall meet the special requirements table of PPP-B-636.

5.3.3 Overseas Shipment.- Night vision training aids packaged as specified in 5.2.2 shall be packed in export-type shipping containers conforming to PPP-B-585, PPP-B-601, PPP-B-591, PPP-B-621, or MIL-B-107. Insofar as practical, exterior containers shall be of uniform shape and size, of minimum cube and tare consistent with the protection required, and shall contain identical quantities. The gross weight of each pack shall be limited to approximately 200 pounds. Strapping and closure shall be in accordance with the appendix of the applicable container specification.

5.4 Physical Protection.- Cushioning, blocking, and bracing, as required, shall be in accordance with JAN-P-100, except that for domestic shipment waterproofing requirements for cushioning materials and containers shall be waived. The drop tests of JAN-P-100 shall be waived when packaging and packing are for immediate use or when the drop tests of MIL-P-116 are applicable.

5.5 Marking of Shipments.- Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

Training Aid, Night Vision, Type MP-1

6. NOTES

6.1 Intended Use.- The Type MP-1 training aid covered by this specification is intended for use in night vision indoctrination training for pilots, observers, and aircrew members.

6.2 Ordering Data.- Invitations for bids and the contracts and purchase orders should include the conditions and requirements for the following:

6.2.1 Preproduction Test Sample.- One training aid should be required as a preproduction sample to be subjected to preproduction tests by the contractor to determine compliance with the requirements of this specification and one training aid should be required as the preproduction sample to be submitted to the procuring activity.

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6.2.2 Procurement documents should specify type of packaging and packing required. (See 5.2 and 5.3)

6.2.3 Drawings, Parts Breakdown Lists, and Photographs.- Drawings, parts breakdown lists and photographs of the training aid should be provided for evaluation prior to preproduction inspection tests. The nature of the parts as related to the complete assembly should be adequately portrayed.

Patent Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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